

# MA2047 Algebra och diskret matematik

Något om grundläggande algebra

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27 augusti 2025

# Faktorisering

Konjugat- och kvadreringsreglerna:

- $(a + b)(a - b) = a^2 - ab + ba - b^2 = a^2 - b^2$
- $(a \pm b)^2 = (a \pm b)(a \pm b) = a^2 \pm ab \pm ba + b^2 = a^2 \pm 2ab + b^2$

## Exempel 1

Faktoriser uttrycket

a)  $x^3 + x^2y - 4xy^2 - 4y^3$

b)  $x^4 + 4x^3y - 16xy^3 - 16y^4$

$$\begin{aligned} \text{a) } x^3 + x^2y - 4xy^2 - 4y^3 &= x^2(x+y) - 4y^2(x+y) = (x+y)(x^2 - 4y^2) \\ &= \underline{\underline{(x+y)(x+2y)(x-2y)}} \end{aligned}$$

$$\begin{aligned} \text{b) } x^4 + 4x^3y - 16xy^3 - 16y^4 &= (x^2 + 4y^2)(x^2 - 4y^2) + 4xy(x^2 - 4y^2) \\ &= (x^2 - 4y^2)(x^2 + 4y^2 + 4xy) \\ &= (x+2y)(x-2y)(x+2y)^2 = \underline{\underline{(x-2y)(x+2y)^3}} \end{aligned}$$

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# Fler ekvationer

## Exempel 3

Lös ekvationen

a)  $x^3 - 2x^2 - 3x = 0$

b)  $x = 2\sqrt{x} + 3$

c)  $x^4 - 2x^2 - 3 = 0$

a)  $x^3 - 2x^2 - 3x = x(x^2 - 2x - 3) = 0$

$\Leftrightarrow \underline{x_1 = 0}$  eller  $x^2 - 2x - 3 = 0$

$\Leftrightarrow x = 1 \pm \sqrt{1^2 + 3} = 1 \pm 2$

$\Leftrightarrow \underline{x_2 = 3}, \underline{x_3 = -1}$

b) sätt  $t = \sqrt{x}$ :

$x - 2\sqrt{x} - 3 = t^2 - 2t - 3 = 0$

$\Leftrightarrow t_1 = \sqrt{x_1} = 3 \Leftrightarrow \underline{x_1 = 9}$

$t_2 = \sqrt{x_2} = -1$  saknar reella lösningar!

c) sätt  $t = x^2$ :

$x^4 - 2x^2 - 3 = t^2 - 2t - 3 = 0$

$\Leftrightarrow t_1 = 3, t_2 = -1$

•  $t_1 = x_1^2 = 3 \Leftrightarrow \underline{x = \pm\sqrt{3}}$

•  $t_2 = x_2^2 = -1$  saknar reella lösningar!

## Exempel 4

Lös ekvationen

a)  $\sqrt{2x+3} = x$

b)  $\sqrt{2x+3} = \sqrt{x+1}$

a)  $\sqrt{2x+3} = x$

$$\Rightarrow 2x+3 = x^2$$

$$\Leftrightarrow x^2 - 2x - 3 = 0$$

$$\Leftrightarrow x = 1 \pm \sqrt{1^2 + 3} = 1 \pm 2$$

$$\Leftrightarrow x_1 = 3, \quad x_2 = -1$$

Falsk rot!

Svar:  $\sqrt{2x+3} = x \Leftrightarrow x = 3$

b)  $\sqrt{2x+3} = \sqrt{x+1}$

$$\Rightarrow 2x+3 = x+1$$

$$\Leftrightarrow x = -2 \leftarrow \text{Falsk rot!}$$

$\therefore$  Ekvationen saknar reella lösningar!

## Exempel 5

Lös olikheten  $\frac{2x-1}{x+3} < 1$ .

$$\frac{2x-1}{x+3} < 1$$

$$\Leftrightarrow \underline{2x-1 < 1 \cdot (x+3)} \quad \checkmark$$

$$\Leftrightarrow x < 4$$

Kontroll:  $x = -4 < 4$

$$\frac{2(-4)-1}{-4+3} = \frac{-9}{-1} = 9 < 1 \quad ?!?$$

OBS!  $3 > 2 \Rightarrow 2 \cdot 3 > 2 \cdot 2$

$3 > 2 \not\Rightarrow (-2) \cdot 3 > (-2) \cdot 2$

Gör så här istället:

$$\frac{2x-1}{x+3} < 1 \Leftrightarrow \frac{2x-1}{x+3} - 1 < 0$$

$$\Leftrightarrow \frac{2x-1}{x+3} - \frac{x+3}{x+3} = \frac{2x-1-(x+3)}{x+3} = \frac{x-4}{x+3} < 0$$

Teckenstudium:

|                   | -3 | 4       |                                     |
|-------------------|----|---------|-------------------------------------|
| $x-4$             | -  | 0       | +                                   |
| $x+3$             | -  | 0       | +                                   |
| $\frac{x-4}{x+3}$ | +  | *       | 0                                   |
|                   |    | ↑       |                                     |
|                   |    | Ej def. |                                     |
|                   |    |         | $\therefore \underline{-3 < x < 4}$ |

## Exempel 7

Lös olikheten  $\frac{x-1}{x+2} \leq \frac{x+3}{x-4}$ .

$$\frac{x-1}{x+2} \leq \frac{x+3}{x-4} \Leftrightarrow \frac{x+3}{x-4} - \frac{x-1}{x+2} \geq 0$$

$$\begin{aligned} \Leftrightarrow & \frac{(x+3)(x+2) - (x-1)(x-4)}{(x-4)(x+2)} \\ = & \frac{x^2 + 2x + 3x + 6 - (x^2 - 4x - x + 4)}{(x-4)(x+2)} \\ = & \frac{10x + 2}{(x-4)(x+2)} = \frac{2(5x+1)}{(x-4)(x+2)} \geq 0 \end{aligned}$$

Teckenstudium:

|        | -2 | $-\frac{1}{5}$ | 4 |   |   |   |
|--------|----|----------------|---|---|---|---|
| $5x+1$ | -  | -              | 0 | + | + | + |
| $x-4$  | -  | -              | - | - | 0 | + |
| $x+2$  | -  | 0              | + | + | + | + |
| tot    | -  | *              | + | 0 | - | * |

$$\therefore \underline{\underline{-2 < x \leq -\frac{1}{5} \text{ eller } x > 4}}$$

# Ekvationer och olikheter i Mathematica

- Exempel 4a:

`Solve[Sqrt[2x+3]==x,x]`

- Exempel 6:

`Reduce[(x-1)/(x+2)<=(x+3)/(x-4),x]`